



Department of Computer Science & Engineering
Pundra University of Science & Technology

Final Year Project/Thesis Preliminary Proposal Form

Student Information:

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ID:	0322320105101032	Semester:	7th
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Academic Record:

Semester	Credit Earned	CGPA	Cumulative CGPA
1 st	21.00	3.23	3.23
2 nd	22.00	3.13	3.18
3 rd	21.00	3.19	3.18
4 th	21.00	3.44	3.25
5 th	16.00	3.25	3.24
6 th	20.00	3.19	3.23
Total	121.00		
Overall CGPA:	3.23		

Project/Thesis Details (Tentative):

Title:	Predicting Customer Purchase Intention in E-Commerce: A Machine Learning Approach Using Data from Bangladesh
Field:	Machine Learning & E-Commerce Analytics
Type:	☒ Individual ☐ Group

Group Members (if applicable):

Sl.	Name	ID
1.		
2.		

Extended Abstract:

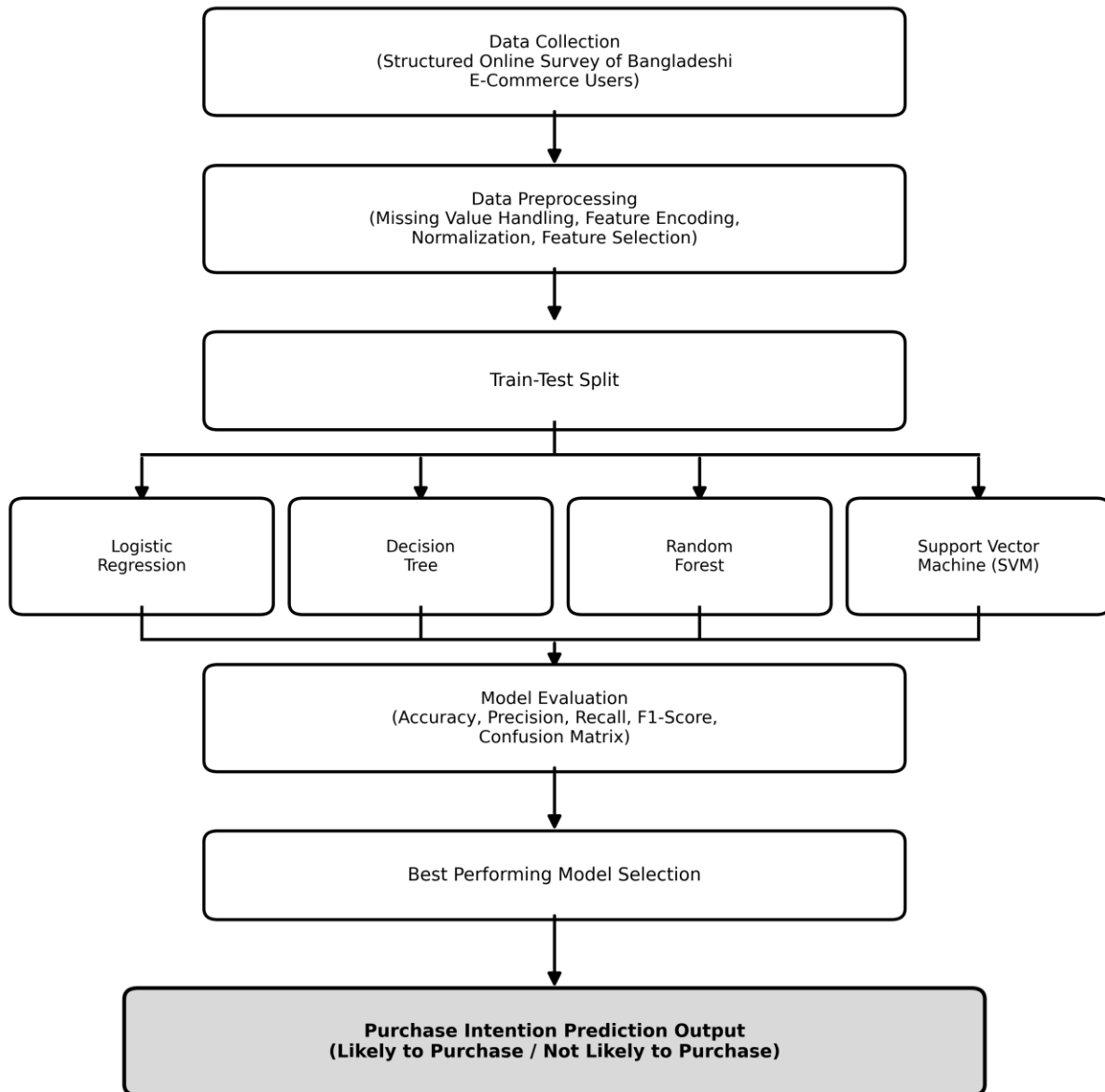
The rapid growth of e-commerce in Bangladesh, driven by increasing internet penetration and smartphone adoption, has transformed consumer shopping behavior. However, despite this growth, online retailers face significant challenges in understanding and predicting which visitors are likely to convert into actual buyers. Purchase intention, defined as a consumer's willingness to buy a product or service, is influenced by numerous factors including browsing behavior, product reviews, pricing, trust, website usability, and demographic characteristics. Traditional statistical approaches often fail to capture the complex, non-linear relationships between these factors and consumer decision-making, resulting in inaccurate predictions and missed business opportunities. This thesis addresses this gap by proposing a machine learning-based framework to predict customer purchase intention using data collected from Bangladeshi online shoppers, thereby providing actionable insights for e-commerce businesses operating in this emerging market.

The proposed methodology follows a structured pipeline consisting of four major phases: data collection, data preprocessing, model development, and evaluation. Data will be gathered through a structured online survey targeting active e-commerce users in Bangladesh, capturing variables such as browsing patterns, trust indicators, payment preferences, and demographic details, alongside behavioral session-level data where applicable. The collected data will undergo preprocessing steps including handling missing values, feature encoding, normalization, and feature selection to ensure model readiness. Multiple supervised machine learning algorithms—including Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine—will be trained and compared to identify the best-performing model based on standard evaluation metrics such as accuracy, precision, recall, and F1-score. A system architecture diagram will illustrate the complete workflow, from data acquisition through preprocessing, model training, and final prediction output, ensuring transparency and reproducibility of the research process.

The expected outcome of this study is a validated predictive model capable of accurately classifying customer purchase intention based on identified behavioral and demographic features, along with insights into which factors most strongly influence online purchasing decisions among Bangladeshi consumers. This research is expected to contribute both theoretically, by extending existing purchase intention literature to the Bangladeshi context, and practically, by offering data-driven guidance to e-commerce businesses.

The findings of this study will primarily benefit e-commerce platform owners, digital marketers, and business strategists seeking to optimize customer targeting, personalize marketing campaigns, and improve conversion rates. Additionally, this research will serve as a foundational reference for future academic studies exploring consumer behavior analytics and machine learning applications within emerging South Asian e-commerce markets.

System Architecture: Predicting E-Commerce Purchase Intention



Student Signature